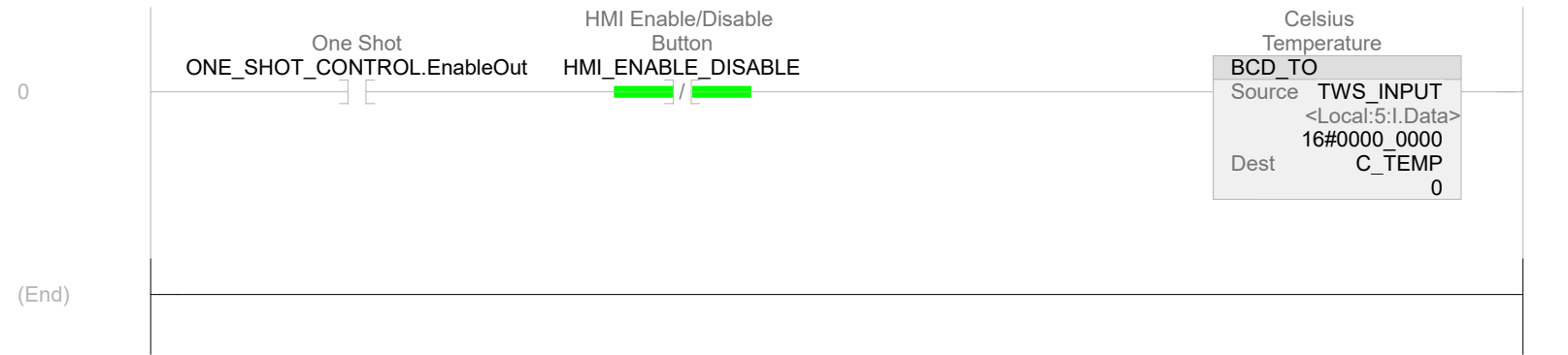




```
1 //Convert a Celsius temperature to a Fahrenheit temperature.
2
3 //Input Parameter
4 SBR(C_TEMP_CONVERT);
5
6 F_TEMP_CONVERT := (C_TEMP_CONVERT * 9) / 5 + 32;
7
8 //Return Parameter
9 RET(F_TEMP_CONVERT);
```

```
1 //Select Case to set range indicator.
2 Case F_TEMP of
3     32..49:  LL_INDC := 1;
4             L_INDC := 0;
5             M_INDC := 0;
6             H_INDC := 0;
7             HH_INDC := 0;
8
9     50..99:  LL_INDC := 0;
10            L_INDC := 1;
11            M_INDC := 0;
12            H_INDC := 0;
13            HH_INDC := 0;
14
15    100..149: LL_INDC := 0;
16            L_INDC := 0;
17            M_INDC := 1;
18            H_INDC := 0;
19            HH_INDC := 0;
20
21    150..199: LL_INDC := 0;
22            L_INDC := 0;
23            M_INDC := 0;
24            H_INDC := 1;
25            HH_INDC := 0;
26
27
28    200..212: LL_INDC := 0;
29            L_INDC := 0;
30            M_INDC := 0;
31            H_INDC := 0;
32            HH_INDC := 1;
33
34
35 end_case;
```

```
1 //One shot for CONVERT PUSHBUTTON or HMI CONVERT PUSHBUTTON.
2 ONE_SHOT_CONTROL.InputBit := HMI_CONVERT_PB OR CONVERT_PB ;
3 OSRI(ONE_SHOT_CONTROL);
4
5 //Subroutine call if Thumb Wheel Switches are enabled.
6 If NOT HMI_ENABLE_DISABLE & ONE_SHOT_CONTROL.OutputBit then
7 JSR(ConvertBCDRoutine);
8 end_if;
9
10 /*If no previous Errors, check for Duplicate or Over Values and set alarms.*/
11 If ONE_SHOT_CONTROL.OutputBit & NOT ERROR_DUP_VAL & NOT ERROR_OVER_VAL then
12     //Check last and current
13     ERROR_DUP_VAL := C_TEMP = C_TEMP_STORAGE;
14     //Check over value
15     ERROR_OVER_VAL := C_TEMP > 100;
16 end_if;
17
18
19 //Error RESET PUSHBUTTON or HMI RESET PUSHBUTTON and clearing alarms.
20 If RESET_PB OR HMI_RESET_PB then
21     ERROR_DUP_VAL := 0;
22     ERROR_OVER_VAL := 0;
23 end_if;
24
25 /* If no errors, store the celsius value and call the conversion
26 routine with the Celsius temperature to return the Fahrenheit
27 temperature. */
28 If ONE_SHOT_CONTROL.OutputBit & NOT ERROR_DUP_VAL & NOT ERROR_OVER_VAL then
29     C_TEMP_STORAGE := C_TEMP;
30     JSR(ConvertRoutine,1,C_TEMP,F_TEMP);
31 end_if;
32
```

```
1
2 //An unconditional Jump to Subroutine to InputControlRoutine.
3 //An unconditional Jump to Subroutine to IndicatorsRoutine
4 JSR(InputControlRoutine);
5 JSR(IndicatorsRoutine);
6
7
```

|                            |   |
|----------------------------|---|
| <b>Convert</b>             |   |
| <b>MainTask</b>            |   |
| <b>MainProgram</b>         |   |
| <b>ConvertBCDRoutine</b>   |   |
| Ladder Diagram .....       | 1 |
| <b>ConvertRoutine</b>      |   |
| Structured Text .....      | 2 |
| <b>IndicatorsRoutine</b>   |   |
| Structured Text .....      | 3 |
| <b>InputControlRoutine</b> |   |
| Structured Text .....      | 4 |
| <b>TextRoutine</b>         |   |
| Structured Text .....      | 5 |